



Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Image Processing

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram explain Single image sensor, how it can be used in Sensor Strip and Sensor Array. (08 Marks)
- b. Explain basic concept of Sampling and Quantization with reference to Digital Image. (07 Marks)
- c. Calculate the photon energy for visible light for given wavelength range 400 nm to 750 nm. [Plank's constant, $h = 6.63 \times 10^{-34}$ Js, $C = 3 \times 10^8$ m/s] (05 Marks)

OR

- 2 a. Explain the Brightness Adaption and Discrimination. (07 Marks)
- b. Explain the Neighbour pixel basic relationship in Digital Images with adjacency connectivity, Regions and Boundaries. (08 Marks)
- c. Given two pixels P and Q with coordinate positions (-2, -2) and (3, 4) respectively, calculate the distance measure D_e , D_4 , D_8 . (05 Marks)

Module-2

- 3 a. Define 2-D orthogonal and unitary transform. (06 Marks)
- b. For given orthogonal matrix A and an image u obtain unitary transform.
Given $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $u = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (08 Marks)
- c. Define the properties of unitary transform. (06 Marks)

OR

- 4 a. Define 2-D DFT and its properties. (06 Marks)
- b. Define cosine transform and its properties. (06 Marks)
- c. Calculate Haar transform for $N = 4$

$$\text{Given Haar function } H_a(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} & , \frac{q-1}{2^p} \leq z < \frac{q-0.5}{2^p} \\ -2^{p/2} & , \frac{q-0.5}{2^p} \leq z < \frac{q}{2^p} \\ 0 & , \text{ else} \end{cases}$$

$$n = \log_2 N$$

$$p = 0 \text{ to } n-1$$

$$q \text{ range between } 1 \leq q \leq 2^p$$

$$k = 2^p + q - 1$$

$$z = 0, 1/4, 2/4, 3/4$$

(08 Marks)

Module-3

- 5 a. With necessary graph and equation explain
 i) Image Negative
 ii) Power law transformation
 iii) Intensity level slicing
 b. Compute Histogram equalization for given data:

(06 Marks)

Table 5(b)

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

for 3 bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) with intensity distribution shown in Table 5(b). Intensity level are integer in range $[0, L-1] = [0, 7]$

(08 Marks)

- c. With an example for 2-bit image of size 5×5 define the sample mean, sample variance with equation.

(06 Marks)

OR

- 6 a. Explain with example fundamentals of Spatial Filtering for spatial correlation and convolution for 1-D and 2-D filter. (08 Marks)
 b. Using 1st order derivative Image Sharpening (the Gradient) define:
 i) Robert's cross gradient operation
 ii) Sobel's operators (for 3×3 region) (06 Marks)
 c. Define smoothing spatial filters with brief note:
 i) Linear Filters
 ii) Order Statistic Filter (06 Marks)

Module-4

- 7 a. With neat block diagram of Homomorphic system, derive Homomorphic filtering approach for Image Enhancement. (08 Marks)
 b. Define sharpening of images in frequency domain using
 i) Ideal High Pass Filter
 ii) Butterworth High Pass Filter
 iii) Gaussian High Pass Filter (06 Marks)
 c. Give Frequency domain filtering necessary steps followed. (06 Marks)

OR

- 8 a. Define pseudo color image processing with intensity slicing and intensity to color transformation. (06 Marks)
 b. Based on Hardware oriented models classify different color model given color conversion for RGB to HIS and vice versa with relevant equation. (08 Marks)
 c. With color fundamentals for primary and secondary colors. (06 Marks)

Module-5

- 9 a. Write brief note on restoration in presence of only noise using
 i) Mean filter ii) Order statistic filter iii) Adaptive filter (08 Marks)
 b. Discuss some of the important noise probability density functions. (06 Marks)
 c. With help of block diagram give details of Degradation / Restoration process. (06 Marks)

OR

- 10 a. In digital images discuss about Inverse Filtering. (06 Marks)
 b. Explain minimum mean square error (Wiener Filter) in Digital Image Processing. (08 Marks)
 c. Discuss periodic noise reduction by frequency domain filtering. (06 Marks)

CMRIT LIBRARY
 BANGALORE - 560 037
